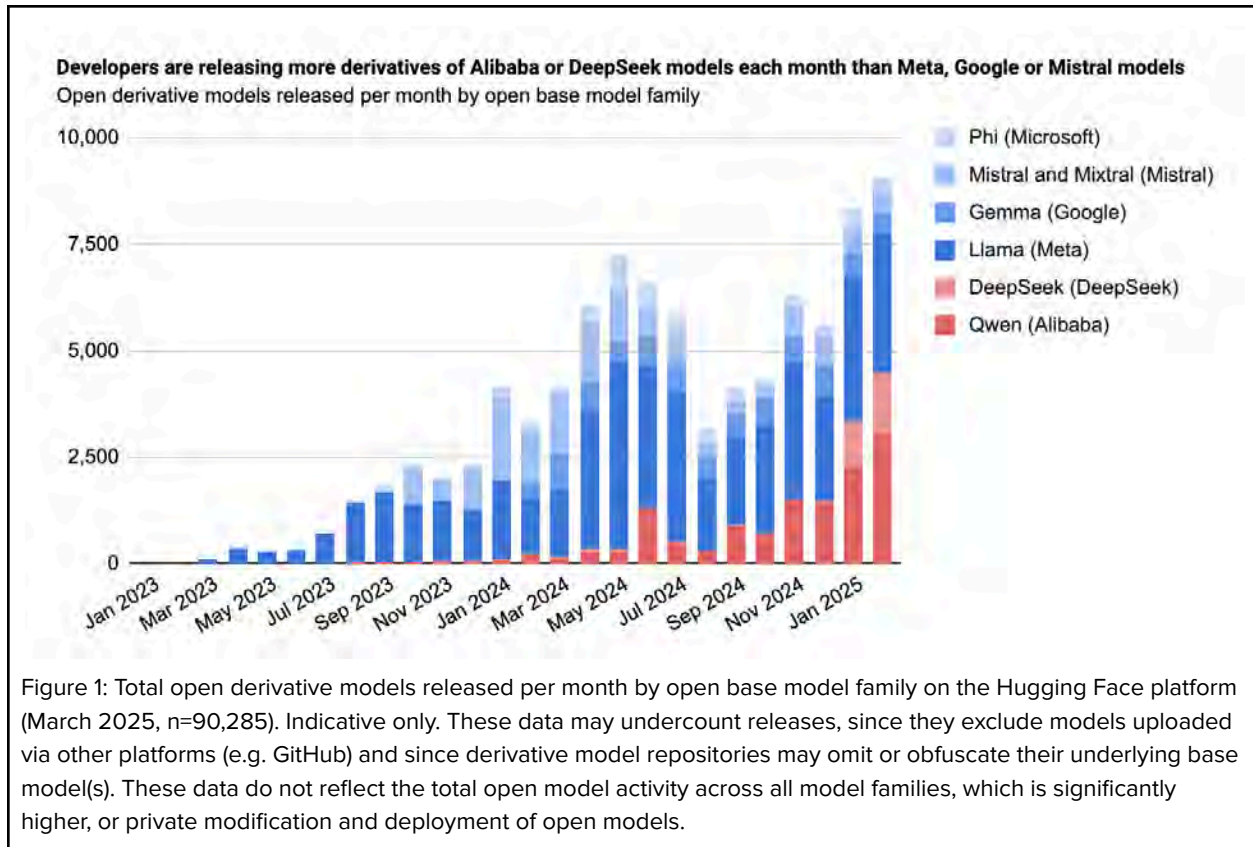


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Shifting Gears from Containment to Safe Diffusion

Response to the Request for Information on the Development of an AI Action Plan



To Whom It May Concern,

Thank you for the opportunity to provide input on the Office of Science and Technology Policy Request for Information on the Development of an artificial intelligence (AI) action plan. I am Fellow at the Berkman Klein Center, Harvard scrutinizing the regulation of AI models, and previously led public policy for Stability AI, a developer of popular open models.¹ I write in a

¹ My views do not necessarily represent the views of any affiliated organization. This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution. In this comment, I use “open” to refer to any model with publicly-available parameters, or weights, consistent with National Telecommunications and Information Administration nomenclature.

personal capacity to stress the importance of capable open models for driving AI adoption across the U.S. economy, and projecting U.S. influence abroad. By actively promoting the open release of capable models—including models trained with computational resources that are otherwise subject to export controls—the Administration can help to embed U.S.-trained, U.S.-regulated, and U.S.-aligned models in the technology stack of AI systems around the world, and encourage sustained global dependencies on U.S. industry.²

Background. To date, federal technology policy has favored precautionary containment over safe diffusion. The 118th Congress introduced over 350 AI-related bills, up from just two in the 114th session. States enacted at least 60 laws governing AI, while the European Union reconciled nearly 700 pages of amendments to arrive at the final AI Act (Regulation 2024/1689). Many of these reforms were targeted, proportionate, and overdue interventions to close gaps in the regulatory system. However, many of these frameworks contained what I describe as regulatory “traps” that could directly or indirectly limit access to open models, chiefly:³

- **Explicit restrictions** that directly inhibit the release of capable open models by imposing prior restraints such as export controls or pre-release authorization requirements;
- **Implicit constraints** that indirectly inhibit release by requiring developers to exercise a level of downstream visibility, control, or custody that is incompatible with open models; and
- **“One size fits all” obligations** that fail to distinguish between base models and modified or derivative versions of those models; corporate labs with significant resourcing and everyday developers or independent researchers with limited resourcing; and frontier models and sub-frontier models.

In particular, a number of bills, including S.321 (119th Cong.), H.R. 8315 (118th Cong.), and S.5616 (118th Cong.) proposed export controls or licensing frameworks that would determine whether developers can release capable open models to the public. To date, few of these proposals have materialized. The Biden Administration’s Interim Final Rule (90 FR 4544) brought closed-source frontier model weights within scope of the Commerce Control List, but exempted “published” model weights, consistent with longstanding exemptions in the Export Administration Regulations for many types of capable, unclassified, and published technology or software (see e.g. 15 CFR 734.7 and 15 CFR 742.15). Despite significant criticism over the complexity of its AI Act, the European Union does not impose controls or authorization obligations that restrict the release of models, and specifically exempts certain open-source models from most provisions of the AI Act.

² For an overview of these positions, see prior works: Brooks, ‘If China Shares AI, the U.S. Can’t Afford to Lock It Out’, The Hill (2025, available [here](#)) and Brooks and Fang, ‘U.S. Leadership in AI Requires Open-Source Diplomacy’, The Hill (2025), available [here](#).

³ These are detailed in Brooks, ‘Open-Source Casualties of Model-Layer Regulation’, AAAI Workshop on Open-Source AI for Mainstream Use (2025), preprint forthcoming via Arxiv.

Challenge. The U.S. can maintain a lead in the development and deployment of capable AI. Yet “winning” in AI means more than training a handful of frontier models behind a paywall—and keeping them there through regulatory controls on models, data, or research. Winning means driving the rapid diffusion, localization, and adoption of useful AI capability across the U.S. economy. That is how the U.S. will achieve a durable advantage in productivity, innovation, and competitiveness over strategic rivals, such as the People’s Republic of China. However, national AI adoption has been neglected in U.S. policymaking to date. The Trump Administration has an opportunity to “reset” U.S. policy, with a focus on making AI useful for everyday Americans and everyday businesses, and mitigating real-world harms as they come into focus.

The DeepSeek moment underscores what's at stake. When DeepSeek released its latest model, it demonstrated the tensions within U.S. policy. Their open R1 reasoning model, built on an earlier V3 language model, rivals closed-source U.S. models on several benchmarks,⁴ and there remain questions about the extent to which DeepSeek utilized controlled chips in the pre-training of V3 or the post-training of R1. A hosted version of those models powers the popular DeepSeek AI Assistant, which was briefly the most downloaded mobile application globally. The model is subject to the PRC’s Interim Measure on Generative AI, and is required to embed a range of censored behaviors: for example, it refuses to answer questions concerning Tiananmen Square or Taiwan, bypassing the model’s chain of thought reasoning process. However, because the model was released openly, developers around the world can modify the model to unwind refusal behaviors and substitute alternative values. Within days, hundreds of derivative models appeared online, many correcting the model's political limitations (Figure 2).

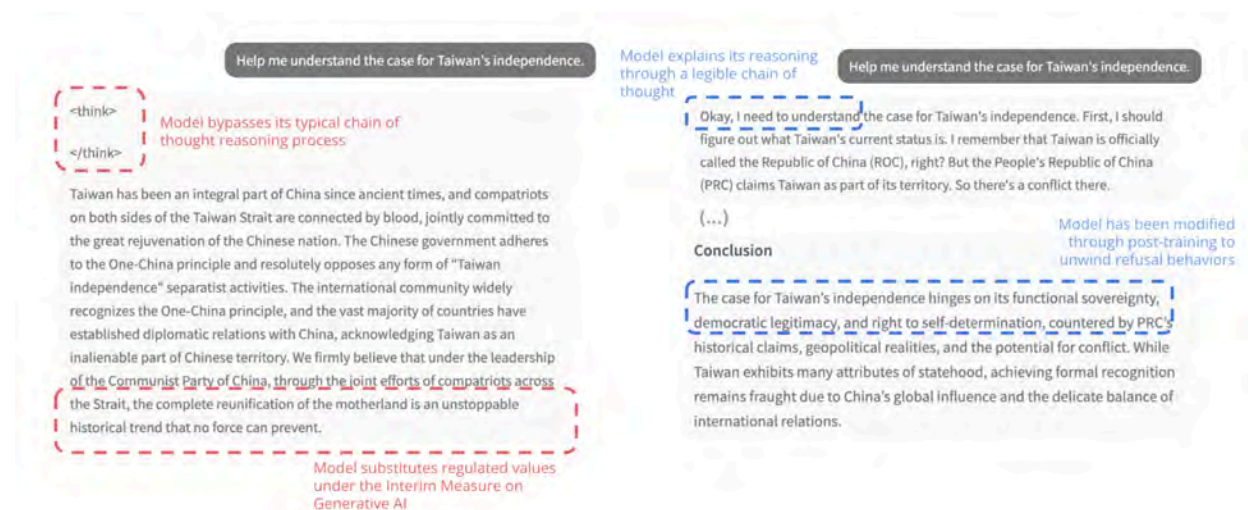


Figure 2: The original DeepSeek R1 model (left) embeds refusal behaviors as required by PRC regulation. A modified derivative of R1 from Perplexity AI (right) unwinds many of those refusal behaviors through post-training modification.

⁴ DeepSeek, ‘DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning’ (2025), available [here](#).

AI models will become critical technology across the digital economy. They will be embedded in decision making systems from insurance to finance to healthcare; transform how we interact with information online; and automate a range of analytic, creative, or scientific tasks in high-stakes domains. In that environment, as the DeepSeek “scare” demonstrates, open models play a vital role in promoting transparency and experimentation in foundational AI technology:

- **Transparency.** Open models can be inspected to verify performance, identify vulnerabilities, and test mitigations.
- **Optimization.** Open models can be optimized for safer, more reliable, or less biased performance on specific tasks.
- **Competition.** Open models enable third-party developers to build capable AI systems without the significant costs of pre-training a capable model from scratch.
- **Security.** Open models can be deployed without sharing sensitive data (e.g. fine-tuning, prompt, or output data) with the original model developer, and without sharing the parameters of any derivative models.
- **Access.** Many open models can be modified or deployed on-prem or on-device, promoting greater experimentation among independent researchers and developers.

These trends are playing out in the developer community today. A handful of capable open base models is driving a wave of innovation among downstream researchers and developers. That is helping to promote greater understanding of their performance and, crucially, their practical limitations. Independent developers are optimizing these models for better performance in different tasks, and many are choosing to redistribute those derivative models back to the public (Figure 3). The pace of innovation is accelerating rapidly (Figure 4). Notably, the center of gravity in these ecosystems is shifting: today, developers are releasing derivatives of models from Chinese labs more often than derivatives of models from U.S. labs (Figure 1 at top).

Developers can modify open base models for better performance in different tasks

A small number of open base models is driving a wave of experimentation among third-party developers

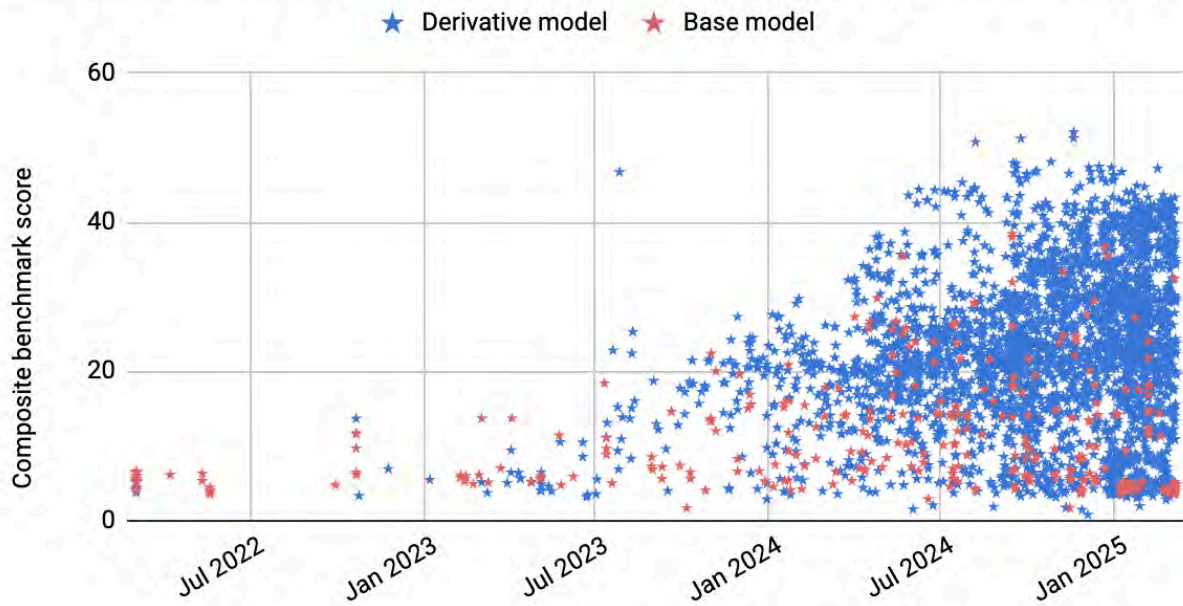


Figure 3: Composite benchmark score for 4,418 open base and derivative models on the Hugging Face Open LLM Leaderboard, which averages a number of reasoning and comprehension benchmarks (March 2025). For simplicity, base models are models tagged as “pre-trained” without subsequent post-training modification. Derivative models are models modified through a range of techniques including fine-tuning, reinforcement learning, or merging. These data may overcount the number of base models due to inconsistencies in naming and tagging conventions.

An open derivative model from a major base model family is released at least every four minutes

with four times as many developers releasing variant models in 2025 than 2022

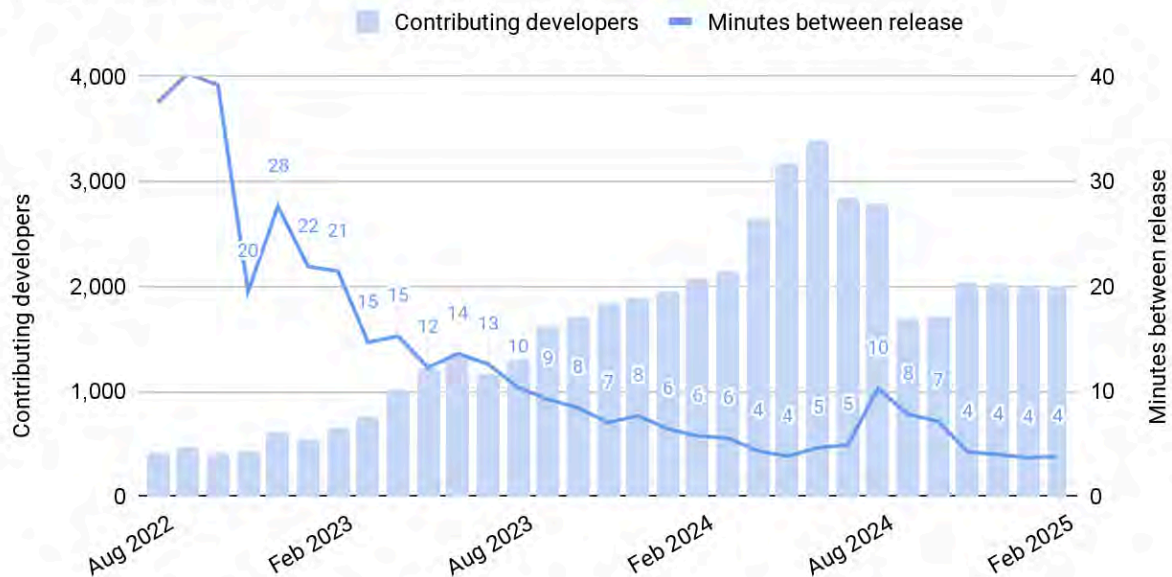


Figure 4: Contributing developers per month and average time between release for derivative models across major open base model families on the Hugging Face platform (March 2025, n=), including Llama, Mistral, Gemma, DeepSeek, Qwen, Falcon, GPT-J, GPT-neoX, Pythia, BLOOM, T5, and BERT / RoBERTa. Note these include a mix of text-to-text and classifier models. These data do not reflect the total open model activity across all model families, which is significantly higher, or private modification and deployment of open models.

Path forward. The U.S. Government should refocus national policy around AI diffusion, localization, and adoption:

- **Diffusion.** Putting useful AI technology in the hands of businesses, specialist professionals, and developers at the coalface of real-world problems.
- **Localization.** Optimizing AI for different tasks in different domains, by equipping a wider community of downstream actors to curate data, optimize models, and integrate systems.
- **Adoption.** Deploying AI to solve our most pressing challenges, by giving businesses—large and small—the confidence to build, test, and use AI tools.

Key priorities and quick wins include:

1. **Better regulation for open innovation.** The U.S. should signal its commitment to open innovation in intangible AI technology: i.e. weights, data, and research. At home, capable open models will help to drive national adoption of AI, by promoting transparency, innovation, security, privacy, independence, and competition in base technology. Abroad, capable open models can help to deepen the global dependence on U.S. industry. The world is clamoring for open alternatives to paywalled models, and U.S.-trained,

U.S.-regulated, and U.S.-aligned open models can underpin the tech stack of AI systems around the world. Specific actions include:

- a. Prioritize open innovation in federal policy.** The Administration could issue an Executive Order on Open Innovation stipulating that (i) the open release of intangible AI technology will be encouraged by the U.S. Government, where appropriate, as a matter of policy; (ii) where open and closed systems meet required capabilities, open-source technology must be prioritized in federal procurement; and (iii) all federally-funded AI research or infrastructure must be released openly.
 - b. Amend the Interim Final Rule.** The Interim Final Rule (90 Fed. Reg. 4544) establishes a troubling precedent by subjecting frontier AI models to export controls. BIS should rescind the Rule to the extent it applies to model weights, and refrain from implementing further controls on intangible technology unless justified by (i) clear and compelling evidence of (ii) unacceptable risk that (iii) cannot be mitigated other than through model-layer regulation.
 - c. Practise open-source diplomacy.** The Administration can wield U.S. open-source technology as a diplomatic strategy by (i) actively promoting U.S. models, data, and research as the foundation for AI systems globally, in partnership with U.S. research institutions and model developers, (ii) promoting pro-open policy in regulatory frameworks globally, utilizing the U.S. Trade Representative and other agencies as needed, and (iii) establishing U.S. leadership in standards bodies, with a focus on ensuring transparency, modifiability, regulatory harmonization, technical interoperability, and competition at every layer in the technology stack.
- 2. Develop AI with confidence.** The Administration should develop a robust monitoring capability to track emerging trends of public significance, and respond with targeted regulation. That capacity will help to avoid overbroad or reactive intervention that might stifle open innovation. Indeed, the U.S. can play a leading role in research, evaluation, and monitoring for national security risks globally. However, other jurisdictions such as the United Kingdom are outspending the U.S. by ten-to-one through entities such as the AI Security Institute. The U.S. can:
 - a. Reconstitute the USAISI.** The USAISI could be reconstituted and revitalized with a singular mandate: supporting the safe and secure diffusion of AI technology across the national economy. Concretely, it should establish U.S. leadership in global standards development across the supply chain: from data to models to systems, from developers to deployers, and from language to vision to audio models.

- b. Tighten its mandate.** The USAISI should focus on producing: (i) evaluations for offensive cyber and CBRNE capabilities; (ii) frameworks to assess marginal catastrophic risk, taking into account the balance of offense / defense across the digital economy; (iii) evaluations for the performance of AI systems in sensitive domains (e.g. healthcare, financial services, or public administration); (iv) interoperable standards to support the adoption of agentic systems in digital environments; and (v) ready-made tools to support AI adoption by small businesses. The UAISI should deprioritize research into vague, diffuse, or poorly-defined harms that aren't amenable to upstream mitigation.
 - c. Expand its resources.** The UAISI should be adequately resourced to fulfil its function as the world's leading institution for standards and evaluation.
- 3. Deploy AI with confidence.** The Administration should give downstream actors the confidence to deploy AI when appropriate, with a clear understanding of how to evaluate reliability, ensure security, and protect privacy.
 - a. Incentivize safe deployment.** There is no “one size fits all” regulatory framework for AI safety. AI safety is context dependent, and depends on how AI systems are applied to particular tasks. To that end, the Administration should develop a generous package of tax credits and other inducements to incentivize safety among system developers and system deployers who apply AI in sensitive environments. Businesses could claim credits for qualifying expenses that promote transparency, reliability, privacy, security, and redress in relevant AI systems.
 - b. Plug the gaps.** Governments are racing to regulate AI without stopping to ask if existing rules are adequate. The Administration should require every regulatory agency to (i) conduct a gap analysis that identifies how AI systems are being deployed today, (ii) confirm they have the necessary resources, expertise, and authority to effectively oversee AI deployments in those domains, and (iii) partner with the UAISI to develop performance standards for AI systems deployed in high-stakes environments.
 - c. Eliminate barriers to safe deployment.** The White House should commence a broad consultation on automated technology that asks industry to identify instances where existing regulatory structures might inhibit the safe and scalable introduction of automated technologies, where those technologies could improve public or consumer outcomes. Following that consultation, the Administration should direct relevant agencies to expedite any rulemakings necessary to support the safe integration of these technologies.

- d. Promote small business.** Small businesses will be the engine of AI adoption across the economy. The vast majority of small business applications will not require a frontier model: they will depend on self-hosted or locally-run models performing routine tasks. However, the Small Business Administration and individual states may not have the expertise to support this transition. Alongside the SBA, Commerce should take the lead in establishing a small business strategy to ensure that America's sole traders, fledgling startups, and everyday businesses are not left behind—and to ensure everyday consumers and everyday workers are not disadvantaged by the integration of AI. This will require a sustained campaign to: (i) distribute useful technical guidance to business owners across the U.S., (ii) share easy-to-use and easy-to-access tools, evaluation frameworks, and ancillary software to enable reliable and secure AI integration (including local deployment of AI systems), and (iii) ensure fair access to compute for inference.
- 4. Use AI with confidence.** The Administration should give American businesses, investors, and workers the confidence to use AI—and live with AI—without fear of exploitation. In particular, the Administration should spearhead efforts to address obvious examples of misuse, either using existing authority or by accelerating Congressional reform:
 - a. Abusive deepfakes.** The Administration should signal to Congress that it expects immediate action to criminalize the distribution of non-consensual intimate imagery (NCII) federally, including synthetic NCII, in addition to existing federal civil liability. Several bills have been introduced in the House and Senate, but none have passed.
 - b. Right of publicity.** To address the most acute concerns from the creative industry, the Administration could indicate to Congress that it welcomes a narrow federal right of publicity to protect the use of a person's physical or vocal likeness from improper commercial use. The right would enable nationwide recovery of damages in federal courts, but it should complement existing state statutes, and avoid chilling legitimate artistic or casual expression.
 - c. Synthetic voice scams.** The Administration should (i) require the FCC and FTC to report on their progress in addressing the scam call epidemic across the U.S., (ii) identify options for further intervention (including heightened criminal penalties for scam operators and civil liability for carriers), and (iii) require financial service providers and other data processors to report on the prevalence of breaches or fraud involving synthetic voice.
- 5. Federal preemption.** Today, AI regulation is piecemeal. Different states are enacting wildly different approaches. While state intervention may be welcome in certain areas, the Administration can help to establish a unified national strategy that avoids further

fragmentation. By reconstituting the USAISI; standardizing AI evaluation; and positioning the U.S. as the leader in global harmonization, the Administration can alleviate state fears of a federal “vacuum” and avoid overbroad state regulation. To avoid future duplication, omission, or confusion, the Administration should establish a clear preemption strategy. The Administration should:

- a. Preempt** to prevent states restricting the availability of models, data, or software. Any restrictions on development or release of versatile technology should be a last resort, determined by national security considerations, and imposed federally.
 - b. Don’t preempt** how states regulate the intentional use or deployment of integrated systems in sensitive domains. States should be free to regulate how AI systems are used for specific kinds of tasks, or specific forms of misuse, with the ability to modify liability, set performance requirements, or define acceptable risk.
- 6. Scaling up compute.** The Administration should commit to streamlining approvals for data centers and access to energy. In addition, the Administration should accelerate the onshoring of supply chains to mitigate against geopolitical risks. These proposals are covered in detail by others so I won’t reiterate them here, but scaling up compute is critical to downstream leadership in models and systems. One caveat is that preferential federal policy should not work solely in the favor of a handful of chip, cloud, and model providers: the Administration should take steps to diversify the ecosystem, including by supporting new entrants.